

1. ITERATIONS OF (α, δ) -SEMI-PROPER FORCINGS

One of the incredible advancements made by Shelah is the concept of *countable support iterations*, which is a technique used to construct iterations of proper forcings while preserving the properness. One of the reasons countable support iterations work is that every proper forcing has the *countable covering property*, stating that every new countable set is covered by a countable set from the ground model. This ensures that the *factor property* holds, which states that a tail-segment of a countable support iteration of proper forcings is again a countable support iteration.

However, when forcing with a poset which is merely *semiproper*, this is no longer the case: For example, although it is semiproper, Prikry forcing adds a countable cofinal set to a former measurable cardinal κ without collapsing it, which clearly implies that this countable set cannot be covered by any $< \kappa$ -sized set from the ground model. Due to this, countable support iterations of semiproper posets can fail to be semiproper and even collapse ω_1 .

The remedy of this is, again, due to Shelah (see [She17, Chapter X]): When iterating semiproper posets, he instead uses *revised countable support iterations*. Roughly speaking, in such iterations, the support of a condition is no longer a countable set of ordinals, but a countable set of *names of ordinals*. As is to be expected, allowing names of ordinals to be used introduces other technical complications. Later on, a more manageable approach to the problem of iterating semiproper posets using Boolean algebras was found by Donder ([Fuc92]) and used e.g. by Jensen ([Jen]). A detailed write-up of this technique was produced by Viale-Audrito-Steila ([VAS14]).

In his book on proper and improper forcing, Shelah proves the iterability of α -properness with countable support and states that (α, ω_1) -semiproperness is iterable with revised countable support by the same proof ([She17, Theorem 1.8 and Theorem 1.9]). Since the technique of revised countable support iterations is still deemed very technical, we provide here a detailed write-up of his result using boolean algebraic iterations.

Now we turn to longer iterations. To do so, we must first introduce some concepts related to the iteration of forcings using Boolean algebras.

The following is well-known:

Proposition 1.1. *Let $\mathbb{P}$ be a poset. Then there is a boolean algebra $\mathbb{B}(\mathbb{P})$ and a dense embedding $\iota: \mathbb{P} \rightarrow \mathbb{B}(\mathbb{P})$.*

Iterating with Boolean algebras requires us to move from the usual notion of iteration (where conditions are functions with a given support) to the following one:

Definition 1.2. Let $(\mathbb{B}_\alpha)_{\alpha < \gamma}$ be a sequence of complete Boolean algebras. We say that $(\mathbb{B}_\alpha)_{\alpha < \gamma}$ is an *iteration* if for each $\alpha < \beta < \gamma$, $\mathbb{B}_\alpha$ is a complete suborder of $\mathbb{B}_\beta$.

As we are working with Boolean algebras, the notions of regular embeddings and projections are dual to each other:

Definition 1.3. Let $(\mathbb{B}_\alpha)_{\alpha < \gamma}$ be an iteration. Let $\mathbb{B} := \bigcup_{\alpha < \gamma} \mathbb{B}_\alpha$ and let $h_\alpha: \mathbb{B} \rightarrow \mathbb{B}_\alpha$ be defined by

$$h_\alpha(b) := \bigwedge \{c \in \mathbb{B}_\alpha \mid b \leq c\}$$

We first record the following easy facts for later:

Lemma 1.4. *Let $(\mathbb{B}_\alpha)_{\alpha < \gamma}$ be an iteration. Let $b \in \bigcup_{\alpha < \gamma} \mathbb{B}_\alpha$ and let $\alpha < \gamma$.*

- (1) *if $b' \leq b$, then $h_\alpha(b') \leq h_\alpha(b)$.*

- (2) $h_\alpha(b) = 0 \leftrightarrow b = 0$;
- (3) $h_\alpha(\bigvee_{i \in I} b_i) = \bigvee_{i \in I} h_\alpha(b_i)$;
- (4) for every $a \in \mathbb{B}_\alpha$, $h_\alpha(b \wedge a) = a \wedge h_\alpha(b)$.

Proof. (1): If $b' \leq b$, then for every c with $b \leq c$, $b' \leq c$. Thus $h_\alpha(b')$ takes a conjunction of a larger set and is thus smaller.

(2): Clearly, if $b = 0$, then $b \leq c$ for all $c \in \mathbb{B}_\alpha$, which implies $h_\alpha(b) = 0$. On the other hand, let $b \neq 0$. Then there is $b' \in \mathbb{B}_\alpha$ such that every $c \leq b'$ in $\mathbb{B}_\alpha$ is compatible with b . We claim that $b' \leq h_\alpha(b)$. Otherwise, there is $c \in \mathbb{B}_\alpha$ with $b \leq c$ and $b' \not\leq c$. In particular, $b' \wedge \neg c \neq 0$. However, the latter condition is below b' and incompatible with c , a contradiction.

(3): By (1), for every $i \in I$, $h_\alpha(b_i) \leq h_\alpha(\bigvee_{i \in I} b_i)$ and thus $\bigvee_{i \in I} h_\alpha(b_i) \leq h_\alpha(\bigvee_{i \in I} b_i)$. On the other hand, let $c \in \mathbb{B}_\alpha$, $c \geq \bigvee_{i \in I} b_i$. Then $c \geq b_i$ for every $i \in I$ and thus $c \geq h_\alpha(b_i)$ for every $i \in I$, which implies $c \geq \bigvee_{i \in I} h_\alpha(b_i)$. Since c was arbitrary,

$$h_\alpha\left(\bigvee_{i \in I} b_i\right) = \bigwedge \left\{ c \in \mathbb{B}_\alpha \mid \bigvee_{i \in I} b_i \leq c \right\} \geq \bigvee_{i \in I} h_\alpha(b_i)$$

(4): We have the following:

$$\begin{aligned} h_\alpha(b \wedge a) &= \bigwedge \{ c \in \mathbb{B}_\alpha \mid b \wedge a \leq c \} \\ &= \bigwedge \{ c \wedge a \mid c \in \mathbb{B}_\alpha, b \leq c \} \\ &= \bigwedge \{ c \in \mathbb{B}_\alpha \mid b \leq c \} \wedge a \\ &= h_\alpha(b) \wedge a \end{aligned}$$

For the second equality, notice that if $b \wedge a \leq c$, $b \leq c \vee \neg a$ and $(c \vee \neg a) \wedge a = c$, so every c with $b \wedge a \leq c$ can be written as $d \wedge a$ for $d \in \mathbb{B}_\alpha$ with $b \leq d$. $\square$

Lemma 1.5. Let $(\mathbb{B}_\alpha)_{\alpha < \gamma}$ be an iteration. Then for every $\alpha < \beta < \gamma$, $h_\alpha = h_\alpha \circ h_\beta$.

Proof. Let $b \in \bigcup_{\alpha < \gamma} \mathbb{B}_\alpha$. By the definition,

$$h_\alpha(h_\beta(b)) = \bigwedge \{ c \in \mathbb{B}_\alpha \mid h_\beta(b) \leq c \} = \bigwedge \left\{ c \in \mathbb{B}_\alpha \mid \bigwedge \{ d \in \mathbb{B}_\beta \mid b \leq d \} \leq c \right\}$$

We prove that $h_\alpha(b) = h_\alpha(h_\beta(b))$.

On one hand, if $h_\beta(b) \leq c$, we have

$$b \leq \bigwedge \{ d \in \mathbb{B}_\beta \mid b \leq d \} = h_\beta(b) \leq c$$

and so $b \leq c$. On the other hand, if $b \leq c$, then $c \in \{ d \in \mathbb{B}_\beta \mid b \leq d \}$ and so

$$h_\beta(b) = \bigwedge \{ d \in \mathbb{B}_\beta \mid b \leq d \} \leq c$$

Therefore,

$$\{ c \in \mathbb{B}_\alpha \mid b \leq c \} = \{ c \in \mathbb{B}_\alpha \mid h_\beta(b) \leq c \}$$

and thus both sets have the same infimum. $\square$

Definition 1.6. Let $\overline{\mathbb{B}} = (\mathbb{B}_\alpha)_{\alpha < \gamma}$ be an iteration.

- (1) A *thread* in $\overline{\mathbb{B}}$ is a function $f \in \prod_{\alpha < \gamma} \mathbb{B}_\alpha$ such that for all $\alpha < \beta < \gamma$, $h_\alpha(f(\beta)) = f(\alpha)$. The collection of all threads is $\mathcal{T}(\overline{\mathbb{B}})$.
- (2) A thread f is *eventually constant* if there is $b \in \bigcup_{\alpha < \gamma} \mathbb{B}_\alpha$ such that for all $\alpha < \gamma$, $f(\alpha) = h_\alpha(b)$. The collection of all eventually constant threads is $\mathcal{C}(\overline{\mathbb{B}})$.

- (3) A thread f is *short* if it is either eventually constant or there is $\alpha < \gamma$ such that

$$f(\alpha) \Vdash_{\mathbb{B}_\alpha} \text{cof}(\gamma) = \omega$$

The collection of all short threads is $\mathcal{S}(\overline{\mathbb{B}})$.

Definition 1.7. Let $\overline{\mathbb{B}} = (\mathbb{B}_\alpha)_{\alpha < \gamma}$ be an iteration and let $f, g \in \mathcal{T}(\overline{\mathbb{B}})$. We let $f \leq g$ if for all $\alpha < \gamma$, $f(\alpha) \leq_{\mathbb{B}_\alpha} g(\alpha)$.

Lemma 1.8. Let $\overline{\mathbb{B}} = (\mathbb{B}_\alpha)_{\alpha < \gamma}$ be an iteration and let $\mathcal{C}(\overline{\mathbb{B}}) \subseteq F \subseteq \mathcal{T}(\overline{\mathbb{B}})$. Assume that

$$\forall \alpha < \gamma \forall f \in F \forall b \in \mathbb{B}_\alpha (b \leq f(\alpha) \rightarrow \exists g \in F (g \leq f \wedge g \leq c(b)))$$

Then F is separative and the mappings $\mathbb{B}_\alpha \rightarrow F$, $b \mapsto c(b)$ are complete embeddings.

Proof. We first prove the separativity. To this end, let $f, g \in F$ and assume that $f \not\leq g$. In particular, there is $\alpha < \gamma$ such that $f(\alpha) \not\leq g(\alpha)$. Since $\mathbb{B}_\alpha$ is a Boolean algebra, it is separative and thus there is $b \in \mathbb{B}_\alpha$ such that $b \leq f(\alpha)$ and $b \perp_{\mathbb{B}_\alpha} g(\alpha)$. By assumption, there is $h \in F$ which extends $c(b)$ and f . In particular, $h(\alpha) = h_\alpha(b) = b$ and thus $h \perp g$ as required.

Now let $\alpha < \gamma$. Let $\iota_\alpha(b) := c(b)$. We prove that ι_α is a complete embedding. It is easy to see that ι_α preserves $\leq$. Let $\mathcal{A} \subseteq \mathbb{B}_\alpha$ be a maximal antichain. We prove that $\iota_\alpha[\mathcal{A}] \subseteq F$ is a maximal antichain. To this end, let $f \in F$. Since $\mathcal{A}$ is a maximal antichain in $\mathbb{B}_\alpha$, there is $c \in \mathcal{A}$ and $d \leq c, f(\alpha)$. By assumption, $\iota_\alpha(d)$ and f are compatible. Thus every element of F is compatible with some element of $\iota_\alpha[\mathcal{A}]$ which is what we wanted to show. $\square$

Corollary 1.9. Let $\overline{\mathbb{B}} = (\mathbb{B}_\alpha)_{\alpha < \gamma}$ be an iteration. Let F be either $\mathcal{T}(\overline{\mathbb{B}})$, $\mathcal{C}(\overline{\mathbb{B}})$ or $\mathcal{S}(\overline{\mathbb{B}})$. Then F is separative and the mappings $\mathbb{B}_\alpha \rightarrow F$, $b \mapsto c(b)$ are complete embeddings.

Proof. We verify that F satisfies the hypothesis of Lemma 1.8.

Let $\alpha < \gamma$, $f \in F$ and $b \in \mathbb{B}_\alpha$ be such that $b \leq f(\alpha)$.

Define $g \in \prod_{\beta < \gamma} \mathbb{B}_\beta$ as follows:

$$g(\beta) := \begin{cases} h_\beta(b) & \beta \leq \alpha \\ f(\beta) \wedge b & \beta > \alpha \end{cases}$$

We first show:

Claim. $g \in \mathcal{T}(\overline{\mathbb{B}})$.

Proof. Let $\beta_0 < \beta_1 < \gamma$. If $\alpha < \beta_0$, we have

$$h_{\beta_0}(g(\beta_1)) = h_{\beta_0}(f(\beta_1) \wedge b) = h_{\beta_0}(f(\beta_1)) \wedge b = f(\beta_0) \wedge b = g(\beta_0)$$

If $\beta_1 < \alpha$, we have

$$h_{\beta_0}(g(\beta_1)) = h_{\beta_0}(h_{\beta_1}(b)) = h_{\beta_0}(b) = g(\beta_0)$$

Lastly, if $\beta_0 \leq \alpha < \beta_1$, we have

$$\begin{aligned} h_{\beta_0}(g(\beta_1)) &= h_{\beta_0}(h_\alpha(g(\beta_1))) \\ &= h_{\beta_0}(h_\alpha(f(\beta_1) \wedge b)) \\ &= h_{\beta_0}(h_\alpha(f(\beta_1)) \wedge b) \\ &= h_{\beta_0}(f(\alpha) \wedge b) \\ &= h_{\beta_0}(b) \\ &= g(\beta_0) \end{aligned}$$

We also show that $g(\beta) \neq 0$ for every $\beta < \gamma$. For $\beta \leq \alpha$, this follows from the assumption that $b \neq 0$. For $\beta > \alpha$, this follows from the fact that $h_\alpha(g(\beta)) = g(\alpha) \neq 0$. $\square$

If $F = \mathcal{T}(\overline{\mathbb{B}})$, the previous claim means that we are done. If $F = \mathcal{C}(\overline{\mathbb{B}})$, there is $d \in \mathbb{B}$ such that $f = c(d)$. In this case, $g = c(d \wedge b)$ and thus in $\mathcal{C}(\overline{\mathbb{B}})$ as well. Lastly, if $F = \mathcal{S}(\overline{\mathbb{B}})$, then either $f \in \mathcal{C}(\overline{\mathbb{B}})$ (and we are done by the previous case) or there is $\beta < \gamma$ such that $f(\beta) \Vdash_{\mathbb{B}_\beta} \text{cof}(\gamma) = \omega$. In this case, it is clear that $g(\beta)$ forces the same so that $g \in \mathcal{S}(\overline{\mathbb{B}})$. $\square$

Definition 1.10. We define the following limits:

$$\begin{aligned} \text{Inv}(\overline{\mathbb{B}}) &:= \mathbb{B}(\mathcal{T}(\overline{\mathbb{B}})) \\ \text{Dir}(\overline{\mathbb{B}}) &:= \mathbb{B}(\mathcal{C}(\overline{\mathbb{B}})) \\ \text{RCS}(\overline{\mathbb{B}}) &:= \mathbb{B}(\mathcal{S}(\overline{\mathbb{B}})) \end{aligned}$$

Definition 1.11. Let $(\mathbb{B}_i)_{i < \gamma}$ be an iteration. We say that $(\mathbb{B}_i)_{i < \gamma}$ is an RCS-iteration if for every limit $\delta < \gamma$, $\mathbb{B}_\delta = \text{RCS}((\mathbb{B}_i)_{i < \delta})$.

The following is clear:

Theorem 1.12. Let γ be an ordinal and let F be a function on γ . Then there is an RCS-iteration $(\mathbb{B}_i)_{i < \gamma+1}$ such that for every $i < \gamma$, $\mathbb{B}_{i+1} = \mathbb{B}_i * F(i)$, provided $F(i)$ is a $\mathbb{B}_i$ -name for a boolean algebra.

The reason behind working with complete Boolean algebras as opposed to simple partial orders is the *factor property*, which roughly states that tails of RCS-iterations are still RCS-iterations.

Definition 1.13. Let $\mathbb{B}$ and $\mathbb{C}$ be complete Boolean algebras such that $\mathbb{B}$ is a complete subalgebra of $\mathbb{C}$. Let G be $\mathbb{B}$ -generic. We let $\mathbb{C}/G$ be the algebra of all equivalence classes of the form $[c]_G$, where $c \sim_G c'$ iff $c' \Delta c \in G$, with the usual operations.

Definition 1.14. Let $\overline{\mathbb{B}} = (\mathbb{B}_\alpha)_{\alpha < \gamma}$ be an iteration. Let $\alpha < \gamma$ and let G be $\mathbb{B}_\alpha$ -generic. In $V[G]$, we define $\overline{\mathbb{B}}/G := (\mathbb{B}_\beta/G)_{\alpha < \beta < \gamma}$.

The following is clear:

Lemma 1.15. Let $\overline{\mathbb{B}} = (\mathbb{B}_\alpha)_{\alpha < \gamma}$ be an iteration. Let $\alpha < \gamma$ and let G be $\mathbb{B}_\alpha$ -generic. Then $\overline{\mathbb{B}}/G$ is an iteration.

Lemma 1.16. Let $\overline{\mathbb{B}} = (\mathbb{B}_\alpha)_{\alpha < \gamma}$ be an iteration. Let $\alpha < \gamma$ and let G be $\mathbb{B}_\alpha$ -generic. Then

$$\text{RCS}(\overline{\mathbb{B}})/G \cong \text{RCS}(\overline{\mathbb{B}}/G)$$

Proof. We will define a dense embedding from $\mathcal{S}(\overline{\mathbb{B}})/G$ into $\mathcal{S}(\overline{\mathbb{B}}/G)$. It then follows that there is a dense embedding from $\mathcal{S}(\overline{\mathbb{B}})/G$ into $\text{RCS}(\overline{\mathbb{B}}/G)$. By the uniqueness of the Boolean completion, we thus obtain an isomorphism of $\text{RCS}(\overline{\mathbb{B}})/G$ and $\text{RCS}(\overline{\mathbb{B}}/G)$.

Let $[f] \in \mathcal{S}(\overline{\mathbb{B}})/G$. Let $\iota(f)$ be a function with domain (α, γ) such that $\iota(f)(\beta) = [f(\beta)]$. It follows that ι is well-defined.

We show that the image of ι is dense. To this end, let $\tau^G \in \mathcal{S}(\overline{\mathbb{B}}/G)$. By the definition of $\mathcal{S}(\overline{\mathbb{B}}/G)$, it follows that either $\tau^G \in \mathcal{C}(\overline{\mathbb{B}}/G)$ or there is $\beta \in (\alpha, \gamma)$ such that $\tau^G(\beta) \Vdash_{\mathbb{B}_\beta/G} \text{cf}(\gamma) = \omega$. We check two cases:

Case A: $\tau^G \in \mathcal{C}(\overline{\mathbb{B}}/G)$. Let $[c]_G \in \mathbb{B}_\beta/G$ for $\beta < \gamma$ be such that $\tau^G = c([c]_G)$. Let f be a function on γ defined by $f(\beta) = [h_\beta(c)]_G = h_\beta([c]_G)$. Then $f \in \mathcal{C}(\overline{\mathbb{B}}/G)$ and $\iota([f]) = \tau^G$, so we are done.

Case B: $\tau^G \in \mathcal{T}(\overline{\mathbb{B}}/G)$. Therefore there is $\beta \in (\alpha, \gamma)$ such that $\tau^G(\beta) \Vdash_{\mathbb{B}_\beta/G} \text{cf}(\gamma) = \omega$. Let $p \in G$ force that $\tau(\beta) \Vdash_{\mathbb{B}_\beta/\mathbb{B}_\alpha} \text{cf}(\gamma) = \omega$. In V , let f be a function on γ defined as follows:

$$f(\delta) := \begin{cases} h_\delta(p) & \delta \leq \alpha \\ \bigvee_{c \in \mathbb{B}_\delta} p \wedge c \wedge [[c]_{\mathbb{B}_\alpha} = \tau(\check{\delta})] & \delta > \alpha \end{cases}$$

We want to show that $f \in \mathcal{T}(\overline{\mathbb{B}})$. To this end, we first show that for every $\delta > \alpha$, $h_\alpha(f(\delta)) = f(\alpha) = p$. Clearly, $h_\alpha(f(\delta)) \leq p$. So assume that $p \not\leq h_\alpha(f(\delta))$. Then there is $r \leq p$ in $\mathbb{B}_\alpha$ which is incompatible with $h_\alpha(f(\delta))$. However, there must be $r' \leq r$ forcing $\tau(\check{\delta}) = [c]_{\mathbb{B}_\alpha}$ for some $c \in \mathbb{B}_\delta$ which means that $r' \wedge c$ is nonzero, $h_\alpha(r' \wedge c) = r' \wedge h_\alpha(c) \leq h_\alpha(f(\delta))$ and $h_\alpha(r' \wedge c) \leq r' \leq r$, a contradiction.

Now let $\alpha < \delta_0 < \delta_1 < \gamma$. We want to show that $h_{\delta_0}(f(\delta_1)) = f(\delta_0)$. We first show $h_{\delta_0}(f(\delta_1)) \leq f(\delta_0)$. To this end, we have that

$$h_{\delta_0}(f(\delta_1)) = \bigvee_{c \in \mathbb{B}_{\delta_1}} h_{\delta_0}(p \wedge c \wedge [[c]_{\mathbb{B}_\alpha} = \tau(\check{\delta}_1)])$$

However, since τ is forced to be in $\mathcal{T}(\overline{\mathbb{B}}/G)$, $[[[c]_{\mathbb{B}_\alpha} = \tau(\check{\delta}_1)]] \leq [[h_{\delta_0}([c]_{\mathbb{B}_\alpha}) = \tau(\check{\delta}_0)]]$. So, in particular,

$$\bigvee_{c \in \mathbb{B}_{\delta_1}} h_{\delta_0}(p \wedge c \wedge [[c]_{\mathbb{B}_\alpha} = \tau(\check{\delta}_1)]) \leq \bigvee_{c \in \mathbb{B}_{\delta_1}} p \wedge h_{\delta_0}(c) \wedge [[h_{\delta_0}(c)]_{\mathbb{B}_\alpha} = \tau(\check{\delta}_0)] = f(\delta_0)$$

On the other hand, suppose that $h_{\delta_0}(f(\delta_1)) \not\leq f(\delta_0)$. In particular, there is $r \leq f(\delta_0)$ which is incompatible with $h_{\delta_0}(f(\delta_1))$. However, we can find $r' \leq h_\alpha(r')$ which decides the value of $\tau(\check{\delta}_1)$ to be equal to some $[c]_{\mathbb{B}_\alpha}$. It follows that r' also decides that $\tau(\check{\delta}_0) = [h_{\delta_0}(c)]_{\mathbb{B}_\alpha}$. However, this implies that $r' \wedge h_{\delta_0}(c)$ is nonzero (since r' and $h_{\delta_0}(c)$ must be compatible) and below r' . Furthermore, $r' \wedge c \leq f(\delta_1)$ and so $h_{\delta_0}(r' \wedge c) = r' \wedge h_{\delta_0}(c)$ witnesses that r and $h_{\delta_0}(f(\delta_1))$ are compatible, a contradiction.

Additionally, it follows that $f(\beta) \Vdash \text{cf}(\gamma) = \omega$, so $f \in \mathcal{S}(\overline{\mathbb{B}})$. In $V[G]$, $[f]_G \in \mathcal{S}(\overline{\mathbb{B}})/G$ and $[f]_G \leq \tau$ by construction, so we are done. $\square$

We now show that suitable limits of (α, ω_1) -semiproper posets of length $\leq \omega_1$ are once again (α, ω_1) -semiproper. The assumption that the length of the limit is $\leq \omega_1$ is crucial at one step in the proof. This assumption is why, in any iteration of semiproper forcing, Levy-collapses are interleaved at every second step to ensure that all limits are either inaccessible or have length $\leq \omega_1$ (modulo reordering).

The proof is quite similar to the proof of the iterability of α -properness ([Abr10, Theorem 5.5])

Lemma 1.17. *Let $\overline{\mathbb{B}} = (\mathbb{B}_i)_{i < \gamma}$ be an iteration where $\gamma \leq \omega_1$ is a limit ordinal and let α be a countable ordinal such that for each $i < j < \gamma$, $\mathbb{B}_i$ forces that $\mathbb{B}_j/\mathbb{B}_i$ is (α, ω_1) -semiproper. Suppose that for each countable $\delta < \gamma$, $\mathbb{B}_\delta = \text{Inv}((\mathbb{B}_i)_{i < \delta})$. Then the following holds:*

- (1) *If $\gamma < \omega_1$, $\text{Inv}(\overline{\mathbb{B}})$ is (α, ω_1) -semiproper;*
- (2) *If $\gamma = \omega_1$, $\text{Dir}(\overline{\mathbb{B}})$ is (α, ω_1) -semiproper.*

Proof. We may assume for simplicity that $\mathbb{B}_0 = 2$ is the trivial Boolean algebra. We prove both statements simultaneously by induction on α . Let $\mathbb{B}_\gamma$ be either $\mathcal{T}(\overline{\mathbb{B}})$ or

$\mathcal{C}(\overline{\mathbb{B}})$ depending on whether $\gamma < \omega_1$ or $\gamma = \omega_1$. Since $\mathbb{B}_\gamma$ is dense in the respective limit, it suffices to show that $\mathbb{B}_\gamma$ is (α, ω_1) -semiproper.

Case A: $\alpha = 1$. So, morally speaking, an α -tower is simply a countable elementary substructure. Let $M \prec H(\Theta)$ be countable with $\mathbb{B}_\gamma \in M$ and let $p \in \mathcal{C}(\overline{\mathbb{B}}_\gamma) \cap M$. We will assume for simplicity that p is the trivial condition. Let $(\tau_n)_{n \in \omega}$ enumerate all $\mathbb{B}_\gamma$ -names for countable ordinals lying in M and let $(\gamma_n)_{n \in \omega}$ be an increasing sequence of ordinals converging to $M \cap \gamma$ such that $\gamma_0 = 0$. By induction on $n < \omega$ we define $p_n \in \mathbb{B}_\gamma$ such that

- (1) $h_{\gamma_n}(p_n)$ is $(M, \mathbb{B}_{\gamma_n})$ -semigeneric;
- (2) it is forced (over $\mathbb{B}_{\gamma_n}$) that $[p_n]_{\mathbb{B}_{\gamma_n}} \in M[\Gamma_{\mathbb{B}_{\gamma_n}}]$;
- (3) for all $k < n$, $p_n \leq p_k$;
- (4) for all $k < n$ and $\alpha \in (\gamma_k, \gamma_n]$, $h_\alpha(p_n) = h_\alpha(p_k)$;
- (5) for all $k < n$, $p_n \Vdash_{\mathbb{B}_\gamma} \tau_n < M[\Gamma_{\mathbb{B}_{\gamma_n}}] \cap \omega_1$.

Let p_0 be trivial. Since $\mathbb{B}_{\gamma_0}$ is trivial, p_0 is $(M, \mathbb{B}_{\gamma_n})$ -semigeneric and it is clear that the other statements hold as well.

So suppose p_n has been defined. For $c \in \mathbb{B}_\gamma$, let $\varphi(c)$ state the following in the forcing language of $\mathbb{B}_{\gamma_n}$:

- (1) $h_{\gamma_n}(c) \leq h_{\gamma_n}(p_n)$;
- (2) $[c]_{\mathbb{B}_{\gamma_n}}$ decides $[\tau_n]_{\mathbb{B}_{\gamma_n}}$ to be an element of $M[\Gamma_{\mathbb{B}_{\gamma_n}}]$;
- (3) $h_{\gamma_{n+1}}([c]_{\mathbb{B}_{\gamma_n}})$ is $(M[\Gamma_{\mathbb{B}_{\gamma_n}}], \mathbb{B}_{\gamma_{n+1}}/\mathbb{B}_{\gamma_n})$ -semigeneric;
- (4) $h_{\gamma_{n+1}}(c)$ forces that $[c]_{\mathbb{B}_{\gamma_{n+1}}} \in M[\Gamma_{\mathbb{B}_{\gamma_{n+1}}}]$

Claim 1. *For every $q \leq h_{\gamma_n}(p_n)$, there is c such that $h_{\gamma_n}(c) \leq q$ and $h_{\gamma_n}(c) \leq [[\varphi(c)]]$.*

Proof. Let $q \leq h_{\gamma_n}(p_n)$. Let G_n be $\mathbb{B}_{\gamma_n}$ -generic containing q . By assumption, $[p_n]_{G_n} \in M[G_n]$. By elementarity, we can find $[a]_{G_n} \leq [p_n]_{G_n}$ in $M[G_n]$ which decides $[\tau_n]_{G_n}$ to be some element of $M[G_n]$. Since $[a]_{G_n} \in M[G_n]$, $[a]_{G_n} \in \mathbb{B}_\delta/G_n$ for some $\delta < M[G_n] \cap \omega_1$. By assumption, $M[G_n] \cap \omega_1 = M \cap \omega_1$. Since $\mathbb{B}_{\gamma_{n+1}}/G_n$ is semiproper, there is a condition $[b]_{G_n} \leq h_{\gamma_{n+1}}([a]_{G_n})$ which is $(M[G_n], \mathbb{B}_{\gamma_{n+1}}/G_n)$ -semigeneric.

Let $[d]_{G_n} := [b]_{G_n} \wedge [a]_{G_n}$. Then $h_{\gamma_{n+1}}([d]_{G_n}) = [b]_{G_n}$ and so $[d]_{G_n}$ is nonzero. In particular, d and q are compatible and so $c := d \wedge q$ is nonzero.

We now have the following: $h_{\gamma_n}(c) \leq q \leq h_{\gamma_n}(p_n)$. $[c]_{G_n}$ decides $[\tau_n]_{G_n}$ to be some element of $M[G_n] \cap \omega_1$, since it extends $[a]_{G_n}$. $h_{\gamma_{n+1}}([c]_{G_n})$ is $(M[G_n], \mathbb{B}_{\gamma_{n+1}}/G_n)$ -semigeneric, since it extends $h_{\gamma_{n+1}}([b]_{G_n})$ and lastly, $h_{\gamma_{n+1}}(c)$ forces that $[c]_{G_{n+1}} = [a]_{G_{n+1}}$ since we only modified a on $\mathbb{B}_{\gamma_{n+1}}$ and in particular, the equivalence class is in $M[\Gamma_{\mathbb{B}_{\gamma_{n+1}}}]$.

Thus, by strengthening c if required, we may assume that $h_{\gamma_n}(c) \leq [[\varphi(c)]]$. $\square$

Now let p_{n+1} be defined as follows:

$$p_{n+1}(\alpha) := \begin{cases} h_{\gamma_n}(p_n) & \alpha \leq \gamma_n \\ \bigvee_{c \in \mathbb{B}_\gamma} h_{\gamma_k}(c) \wedge [[\varphi(c)]] & \alpha \in (\gamma_n, M \cap \omega_1) \end{cases}$$

We show that p_{n+1} is as required. First note that $p_{n+1} \in \mathbb{B}_{M \cap \omega_1}$.

Claim. *For every $\alpha \in (\gamma_n, M \cap \omega_1)$, $h_{\gamma_n}(p_{n+1}(\alpha)) = h_{\gamma_n}(p_n)$.*

Proof. It is clear that $h_{\gamma_n}(p_{n+1}(\alpha)) \leq h_{\gamma_n}(p_n)$ simply by the definition of $\varphi(c)$.

So suppose that $h_{\gamma_n}(p_n) \not\leq h_{\gamma_n}(p_{n+1}(\alpha))$ for some α . By the separativity of $\mathbb{B}_{\gamma_n}$, it follows that there is $q \leq h_{\gamma_n}(p_n)$ which is incompatible with $h_{\gamma_n}(p_{n+1}(\alpha))$. However, by the claim, there is c such that $h_{\gamma_n}(c) \leq q$ and $h_{\gamma_n}(c) \leq [[\varphi(c)]]$. In particular, $h_\alpha(c) \leq p_{n+1}(\alpha)$ and thus $h_{\gamma_n}(c) = h_{\gamma_n}(h_\alpha(c))$ is below both $h_{\gamma_n}(p_{n+1}(\alpha))$ and $h_{\gamma_n}(p_n)$, a contradiction. $\square$

So $p_{n+1} \in \mathbb{B}_\gamma$. Clearly, $h_{\gamma_n}(p_n)$ forces that $[h_{\gamma_{n+1}}(p_{n+1})]_{\mathbb{B}_{\gamma_n}}$ is $(M[\Gamma_{\mathbb{B}_{\gamma_n}}], \mathbb{B}_{\gamma_{n+1}}/\mathbb{B}_{\gamma_n})$ -semigeneric. Since $h_{\gamma_n}(p_n)$ is $(M, \mathbb{B}_{\gamma_n})$ -semigeneric, it follows easily that $h_{\gamma_{n+1}}(p_{n+1})$ is $(M, \mathbb{B}_{\gamma_{n+1}})$ -semigeneric. By the definition of $\varphi(c)$, it follows that it is forced that $[p_{n+1}]_{\mathbb{B}_{\gamma_{n+1}}} \in M[\Gamma_{\mathbb{B}_{\gamma_{n+1}}}]$. Also, p_{n+1} extends p_n by construction and for all $\alpha \leq \gamma_n$, $h_\alpha(p_{n+1}) = h_\alpha(p_n)$ by the definition of p_{n+1} . Lastly, $[h_{\gamma_{n+1}}(p_{n+1})]_{\mathbb{B}_{\gamma_n}}$ is forced to decide τ_n to be an element of $M[\Gamma_{\mathbb{B}_{\gamma_n}}]$ by assumption, so we are done.

Now assume that we have defined $(p_n)_{n \in \omega}$. Let p_ω be defined as follows: For every $\alpha \in (\gamma_n, \gamma_{n+1})$, $p_\omega(\alpha) = h_\alpha(p_{n+1})$. It follows from the inductive hypothesis that $p_\omega \in \mathbb{B}_{M \cap \gamma}$ and thus that $p_\omega \in \mathbb{B}_\gamma$. Furthermore, clearly $p_\omega \leq p_n$ for every $n \in \omega$.

We are done after showing:

Claim. p_ω is $(M, \mathbb{B}_\gamma)$ -semigeneric.

Proof. Let $\tau \in M$ be a $\mathbb{B}_\gamma$ -name for a countable ordinal. Thus there is $n \in \omega$ such that $\tau = \tau_n$. By the inductive hypothesis, p_ω forces that τ_n is below $M[\Gamma_{\mathbb{B}_{\gamma_{n+1}}}] \cap \omega_1$. However, p_ω also extends p_{n+1} and $h_{\gamma_{n+1}}(p_{n+1})$ is $(M, \mathbb{B}_{\gamma_{n+1}})$ -semigeneric and thus it forces that $M[\Gamma_{\mathbb{B}_{\gamma_{n+1}}}] \cap \omega_1 = M \cap \omega_1$. In summary, p_ω forces that τ_n is below $M \cap \omega_1$. As τ was arbitrary, we are done. $\square$

Case B: α is a limit ordinal. Let $(M_i)_{i < \alpha}$ be an α -tower such that $\mathbb{B}_\gamma \in M_0$ and let $p \in \mathbb{B}_\gamma \cap M_0$. We will assume for simplicity that p is the trivial condition. Fix an increasing sequence $(\alpha_n)_{n \in \omega}$ of successor ordinals converging to α . Also fix a sequence $(\gamma_n)_{n \in \omega}$ converging to $\sup(\sup(M_i \cap \omega_1)_{i < \alpha})$ such that $\gamma_0 = 0$ and for every $n \in \omega$, $\gamma_n \in M_{\alpha_n}$.

By induction on $n \in \omega$, we define $p_n \in \mathbb{B}_\gamma$ such that

- (1) p_n is $((M_i)_{i < \alpha_n}, \mathbb{B}_\gamma)$ -semigeneric;
- (2) $h_{\gamma_n}(p_n)$ is $((M_i)_{i < \alpha}, \mathbb{B}_{\gamma_n})$ -semigeneric;
- (3) $[p_n]_{\mathbb{B}_{\gamma_n}} \in M_{\alpha_n}[\Gamma_{\mathbb{B}_{\gamma_n}}]$;
- (4) for all $k < n$, $p_n \leq p_k$;
- (5) for all $k < n$ and $\beta \in (\gamma_k, \gamma_n]$, $h_\beta(p_n) = h_\beta(p_k)$;

To start, note that the trivial condition is $((M_i)_{i < \alpha}, \mathbb{B}_{\gamma_0})$ -semigeneric and lies in M_0 . By the inductive hypothesis applied in M_{α_0} , we can find $p_0 \in M_{\alpha_0}$ which is $((M_i)_{i < \alpha_0}, \mathbb{B}_{\gamma_0})$ -generic. It is clear that p_0 is as required.

Now suppose that p_n has been defined. For $c \in \mathbb{B}_\gamma$, let $\varphi(c)$ state the following in the forcing language of $\mathbb{B}_{\gamma_n}$:

- (1) $h_{\gamma_n}(c) \leq h_{\gamma_n}(p_n)$;
- (2) $[c]_{\mathbb{B}_{\gamma_n}}$ is $((M_i[\Gamma_{\mathbb{B}_{\gamma_n}}])_{\alpha_n \leq i < \alpha_{n+1}}, \mathbb{B}_{\gamma_n})$ -semigeneric;
- (3) $h_{\gamma_{n+1}}([c]_{\mathbb{B}_{\gamma_n}})$ is $((M_i[\Gamma_{\mathbb{B}_{\gamma_n}}])_{\alpha_{n+1} \leq i < \alpha}, \mathbb{B}_{\gamma_{n+1}}/\mathbb{B}_{\gamma_n})$ -semigeneric;
- (4) $h_{\gamma_{n+1}}(c)$ forces that $[c]_{\mathbb{B}_{\gamma_{n+1}}} \in M[\Gamma_{\mathbb{B}_{\gamma_{n+1}}}]$.

Claim. For every $q \leq h_{\gamma_n}(p_n)$, there is c such that $h_{\gamma_n}(c) \leq q$ and $h_{\gamma_n}(c) \leq [\varphi(c)]$.

Proof. Let $q \leq h_{\gamma_n}(p_n)$. Let G_n be $\mathbb{B}_{\gamma_n}$ -generic containing q . By assumption, $[p_n]_{G_n} \in M_{\alpha_n}[G_n]$. By the inductive assumption applied inside of $M_{\alpha_{n+1}}[G_n]$, we can find $[a]_{G_n} \leq [p_n]_{G_n}$ in $M_{\alpha_{n+1}}[G_n]$ which is $((M_i[G_n])_{\alpha_n \leq i < \alpha_{n+1}}, \mathbb{B}_{\gamma_n})$ -semigeneric. By the assumption that $\mathbb{B}_{\gamma_{n+1}}/G_n$ is semiproper, we can find $[b]_{G_n} \leq h_{\gamma_{n+1}}([a]_{G_n})$ which is $((M_i[G_n])_{\alpha_{n+1} \leq i < \alpha}, \mathbb{B}_{\gamma_{n+1}}/G_n)$ -semigeneric.

Let $[d]_{G_n} := [b]_{G_n} \wedge [a]_{G_n}$. Then $h_{\gamma_{n+1}}([d]_{G_n}) = [b]_{G_n}$ and so $[d]_{G_n}$ is nonzero. In particular, d and q are compatible and so $c := d \wedge q$ is nonzero.

It follows just as before that c is as required. $\square$

Now let p_{n+1} be defined as follows:

$$p_{n+1}(\alpha) := \begin{cases} h_\alpha(p_n) & \alpha \leq \gamma_n \\ \bigvee_{c \in \mathbb{B}_\gamma} h_{\gamma_k}(c) \wedge [[\varphi(c)]] & \alpha \in (\gamma_n, M \cap \omega_1) \end{cases}$$

We show that p_{n+1} is as required. Note that, just like before, we have $p_{n+1} \in \mathbb{B}_{M \cap \omega_1}$. The only thing left to show is the semigenericity.

To show that p_{n+1} is $((M_i)_{i < \alpha_{n+1}}, \mathbb{B}_\gamma)$ -semigeneric, note that already p_n is $((M_i)_{i < \alpha_n}, \mathbb{B}_\gamma)$ -semigeneric by the inductive hypothesis. Furthermore, $h_{\gamma_n}(p_n)$, which is $((M_i)_{i < \alpha_n}, \mathbb{B}_{\gamma_n})$ -semigeneric, forces that $[p_{n+1}]_{\mathbb{B}_{\gamma_n}}$ is $((M_i[\Gamma_{\mathbb{B}_{\gamma_n}}])_{\alpha_n \leq i < \alpha_n}, \mathbb{B}_\gamma)$ -semigeneric. Thus, p_{n+1} is $((M_i)_{\alpha_n \leq i < \alpha_{n+1}}, \mathbb{B}_\gamma)$ -semigeneric as required.

To show that $h_{\gamma_{n+1}}(p_{n+1})$ is $((M_i)_{i < \alpha}, \mathbb{B}_{\gamma_{n+1}})$ -semigeneric, we must only show that it is $((M_i)_{\alpha_{n+1} \leq i < \alpha}, \mathbb{B}_{\gamma_{n+1}})$ -semigeneric. To see this, note that $h_{\gamma_n}(p_n)$ is $((M_i)_{\alpha_{n+1} \leq i < \alpha}, \mathbb{B}_{\gamma_n})$ -semigeneric by the inductive assumption and it forces that $h_{\gamma_{n+1}}([p_{n+1}]_{\mathbb{B}_{\gamma_n}})$ is $((M_i[\Gamma_{\mathbb{B}_{\gamma_n}}])_{\alpha_{n+1} \leq i < \alpha}, \mathbb{B}_{\gamma_{n+1}}/\mathbb{B}_{\gamma_{n+1}})$ -semigeneric, which clearly implies the desired statement.

Now let p_ω be defined as follows: For $\alpha \in [\gamma_n, \gamma_{n+1})$, $p_\omega(\alpha) = h_\alpha(p_{n+1})$. It follows that $p_\omega \in \mathbb{B}_\gamma$. Furthermore, for every $n \in \omega$, $p_\omega \leq p_n$ and so, p_ω is $(M_i, \mathbb{B}_\gamma)$ -semigeneric for every $i < \alpha$ as required.

Case C: $\alpha = \alpha' + 1$, α' a successor. Let $(M_i)_{i < \alpha}$ be an α -tower such that $\mathbb{B}_\gamma \in M_0$ and let $p \in \mathbb{B}_\gamma \cap M_0$. By the inductive hypothesis, applied in $M_{\alpha'}$, we can find $p' \in M_{\alpha'}$ which is $((M_i)_{i < \alpha'}, \mathbb{B}_\gamma)$ -semigeneric. By Case A, we can find $p'' \leq p'$ which is $(M_{\alpha'}, \mathbb{B}_\gamma)$ -semigeneric. It follows easily that p'' is $((M_i)_{i < \alpha}, \mathbb{B}_\gamma)$ -semigeneric.

Case D: $\alpha = \alpha' + 1$, α' a limit. Let $(M_i)_{i < \alpha}$ be an α -tower such that $\mathbb{B}_\gamma \in M_0$ and let $p \in \mathbb{B}_\gamma \cap M_0$. By the inductive hypothesis, let $p' \leq p$ be $((M_i)_{i < \alpha'}, \mathbb{B}_\gamma)$ -semigeneric. Since $M_{\alpha'} = \bigcup_{i < \alpha'} M_i$, it follows easily that p' is $((M_i)_{i < \alpha}, \mathbb{B}_\gamma)$ -semigeneric. $\square$

The only possible limit left to consider is one of inaccessible length. By interleaving enough collapses, a theorem of Baumgartner allows us to assume that, in this situation, the forcing has a sufficiently nice chain condition.

Lemma 1.18. *Let $\bar{\mathbb{B}} = (\mathbb{B}_i)_{i < \gamma}$ be an iteration where γ is an inaccessible cardinal such that $|\mathbb{B}_i| < \gamma$ for every $i < \gamma$. Suppose that $\mathbb{B}_\gamma := \text{Dir}(\bar{\mathbb{B}})$ has the γ -cc and that for every $i < \gamma$, $\mathbb{B}_i$ is (α, ω_1) -semiproper. Then $\mathbb{B}_\gamma$ is (α, ω_1) -semiproper.*

Proof. Let $(M_i)_{i < \alpha}$ be an α -tower such that $\mathbb{B}_\gamma \in M_0$ and let $p \in M_0 \cap \mathbb{B}_\gamma$. Let $\delta := \sup_{i < \alpha} \sup(M_i \cap \gamma)$. Then $\delta < \gamma$. Thus, by the inductive assumption, we can find a condition $p' \leq p$ which is $((M_i)_{i < \alpha}, \mathbb{B}_\delta)$ -semigeneric. We will show that p' is $((M_i)_{i < \alpha}, \mathbb{B}_\gamma)$ -semigeneric. To this end, let $i < \alpha$ and let $\tau \in M_i$ be a $\mathbb{B}_\gamma$ -name for a countable ordinal. By elementarity, there is a maximal antichain $\mathcal{A}$ in M_i of conditions deciding τ . Since $\mathbb{B}_\gamma$ has the γ -cc and $\mathbb{B}_\gamma$ is the direct limit, we can apply elementarity to find an ordinal $\eta \in M_i$ such that $\mathcal{A} \subseteq \mathbb{B}_\eta$. In particular, $\eta < \delta$ and τ is equivalent to a $\mathbb{B}_\delta$ -name in M_i . Thus, since p' is $(M_i, \mathbb{B}_\delta)$ -generic, it forces that $\tau \in M_i \cap \omega_1$ as required. $\square$

The following theorem summarizes this section. As before, note that we involve sufficiently many collapses in order to be able to assume that every limit we take has length $\leq \omega_1$.

Theorem 1.19. *Let $\bar{\mathbb{B}} = (\mathbb{B}_i)_{i < \gamma}$ be an iteration. Suppose the following:*

- (1) *For every $i < \gamma$,*

$$\Vdash_{\mathbb{B}_i} \mathbb{B}_{i+1}/\mathbb{B}_i \text{ is } (\alpha, \omega_1)\text{-semiproper}$$

- (2) *For every limit $\delta < \gamma$, $\mathbb{B}_\delta = \text{RCS}((\mathbb{B}_i)_{i < \delta})$;*
- (3) *For every even ordinal $i < \gamma$, $\Vdash_{\mathbb{B}_{i+1}} |2^{\mathbb{B}_i}| \leq \omega_1$.*

Then for every $i < \gamma$, $\mathbb{B}_i$ is (α, ω_1) -semiproper.

Proof. We will show by induction on $i < \gamma$ that for every $j < i$,

$$\Vdash_{\mathbb{B}_j} \mathbb{B}_i/\mathbb{B}_j \text{ is } (\alpha, \omega_1)\text{-semiproper}$$

Case A: $i = i' + 1$ is a successor. Let $j < i$. If $j = i'$, we are done by assumption. Otherwise, let G_j be $\mathbb{B}_j$ -generic. It follows that $\mathbb{B}_i/G_j$ is forcing equivalent to $(\mathbb{B}_{i'}/G_j) * (\mathbb{B}_i/\mathbb{B}_{i'})$. By Lemma ??, $\mathbb{B}_i/G_j$ is (α, ω_1) -semiproper.

Case B: i is either not inaccessible or there is $\delta < i$ such that $|\mathbb{B}_\delta| \geq i$. Let $j < i$. In this case, there must be a cardinal $\delta \in (j, i)$ such that $2^\delta \geq \text{cf}(i)$: If there is $\delta < i$ such that $|\mathbb{B}_\delta| \geq i$, this is clear. Otherwise, i is either singular in which case $\delta = \text{cf}(i)$ works or not a strong limit. It follows that $\Vdash_{\mathbb{B}_\delta} \text{cf}(i) \leq \omega_1$. So let $((M_i)_{i < \alpha})$ be an α -tower and let $p \in M_0 \cap \mathbb{B}_i$.

Case B0: $h_\delta(p) \Vdash \text{cf}(i) = \omega_1$. Let G_j be $\mathbb{B}_j$ -generic containing $h_j(p)$ and let $[a]_{G_j} \leq h_\delta([h]_{G_j})$ be $((M_i[G_j])_{i < \alpha}, \mathbb{B}_\delta/G_j)$ -semigeneric. Let G_δ be $\mathbb{B}_\delta/G_j$ -generic containing $h_\delta(p)$. In $V[G_\delta]$, let $f: \omega_1 \rightarrow i$ be increasing with $f(0) \geq \delta$. It follows that $\mathbb{B}_i/G_\delta$ is forcing equivalent to $\text{Dir}((\mathbb{B}_{f(\alpha)}/G_\delta)_{\alpha < \omega_1})$, since ω_1 is not collapsed. By the inductive assumption we have that for every $\alpha < \beta < \omega_1$,

$$\Vdash_{\mathbb{B}_{f(\alpha)}} \mathbb{B}_{f(\beta)}/\mathbb{B}_{f(\alpha)} \text{ is } (\alpha, \omega_1)\text{-semiproper}$$

By Lemma 1.17, $\mathbb{B}_i/G_\delta$ is (α, ω_1) -semiproper and $[p]_{G_\delta} = [p]_{G_\delta} \in M[G_\delta]$, so we can find an $((M_k[G_\delta])_{k < \alpha}, \mathbb{B}_i/G_\delta)$ -semigeneric condition $[q]_{G_\delta}$ extending $[p]_{G_\delta}$. In particular, q and a are compatible. It follows that $q \wedge a$ is $((M_k[G_j])_{k < \alpha}, \mathbb{B}_i/G_\delta)$ -semigeneric.

Case B1: $h_\delta(p) \nVdash \text{cf}(i) = \omega_1$. We may therefore assume that $h_\delta(p) \Vdash \text{cf}(i) = \omega$. Let G_j be $\mathbb{B}_j$ -generic containing $h_j(p)$ and let $[a]_{G_j} \leq h_\delta([h]_{G_j})$ be $((M_i[G_j])_{i < \alpha}, \mathbb{B}_\delta/G_j)$ -semigeneric. Let G_δ be $\mathbb{B}_\delta$ -generic containing $h_\delta(p)$. In $V[G_\delta]$, let $f: \omega \rightarrow i$ be increasing with $f(0) \geq \delta$. By the Factor Lemma 1.16, it follows that

$$\mathbb{B}_i/G_\delta = \text{RCS}((\mathbb{B}_k)_{k < i})/G_\delta = \text{RCS}((\mathbb{B}_k/G_\delta)_{\delta \leq k < i}) = \text{Inv}((\mathbb{B}_{f(k)})/G_\delta)_{k \in \omega}$$

By the inductive assumption, we have that for every $k_0 < k_1 < \omega$,

$$\Vdash_{\mathbb{B}_{f(k_0)}/G_\delta} \mathbb{B}_{f(k_1)}/G_\delta \text{ is } (\alpha, \omega_1)\text{-semiproper}$$

Therefore, by Lemma 1.17, $\mathbb{B}_i/G_\delta$ is (α, ω_1) -semiproper. Furthermore, $[a]_{G_\delta} = [p]_{G_\delta}$ and so we can find a $((M_k[G_\delta])_{k < \alpha}, \mathbb{B}_i/G_\delta)$ -semigeneric condition extending $[a]_{G_\delta}$. As before, this allows us to find a $((M_k[G_j])_{k < \alpha}, \mathbb{B}_i/G_\delta)$ -semigeneric condition extending $[p]_{G_j}$ as required.

Case C: i is inaccessible and $|\mathbb{B}_\delta| < i$ for every $\delta < i$. Let $j < i$ and let G_j be $\mathbb{B}_j$ -generic. Clearly, there are stationarily many $\delta \in (j, i)$ such that $\text{cf}(\delta) = \omega_1$. In particular, for any such δ ,

$$\mathbb{B}_\delta/G_j = \text{Dir}((\mathbb{B}_k/G_j)_{k < \delta})$$

It therefore follows from a theorem of Baumgartner that $\mathbb{B}_i/G_j$ has the i -cc. Therefore, by Lemma 1.18, $\mathbb{B}_i/G_j$ is (α, ω_1) -semiproper. $\square$

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